

A THEORY OF FINITE REPRESENTATION

Part III

The Geofinite Audit

Axioms, Mathematical Objects and
Functional Symbolic Trajectories

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Geofinitism and Finite Symbolic Mechanics

Foreword

This third volume follows *A Theory of Finite Representation, Part I: Extent, Registration and the Licence to Represent* and *Part II: Finite Reports, Accounts and Operational Insufficiency*. Its purpose is to turn their representational account into an instrument of inquiry. The principal specimens are classical mathematical constructions and the Zermelo–Fraenkel axioms with Choice. The further outcome is a general method for auditing a symbolic construction in relation to the work it is being asked to perform.

The argument proceeds from within Geofinitism and Finite Symbolic Mechanics, abbreviated FSM. Finite representation, Alphonic distinction, uncertainty, accessible provenance and admissibility are the enabling commitments of this inquiry. Classical mathematics is followed under its own declared permissions when it supplies a specimen. Its representation within an FSM report and the adoption of its commitments as those of an FSM object remain distinguishable acts.

This is a complete first working draft, rather than a completed formal calculus. The Greene–Pascal contribution is introduced as a framing of the Functional Symbolic Trajectory, or supporting region of trajectories, needed to hold and use an Alphonic distinction. Questions of minimum support, cost and quantitative comparison remain available for development. The worked cases are stipulated constructions; the physical partition case is an audit scenario, not a record of a newly performed experiment.

The full reconstruction of alternative FSM mathematical objects belongs to the separate Mathematical Objects programme. This volume establishes what such a reconstruction would need to disclose. Its own terminology, formal examples and source choices are therefore included within the audit. The writing follows the revised outline agreed in October 2026, with its prologue and conclusion outside the chapter numbering.

Abstract

A formal object may occupy the beginning of an argument while arriving late in the trajectory that made the argument possible. Axioms require a language, distinguishable inscriptions, conventions, readers and histories of use. Mathematical objects gather constructive permissions into names that can travel through further proofs. Bridge theorems make results portable between representations and fields. The resulting mobility is functional, but a compact endpoint does not necessarily carry the distinctions required by every later continuation.

This monograph develops a Geofinite audit of those relationships. It connects the finite representation of Part I with the reports, ledgers and conditional insufficiency of Part II. ZFC is examined through a declared first-order presentation, with each axiom or schema followed through its formal permission, a finite use, its representational commitments and the questions arising in an FSM translation. Selected mathematical objects, algebra, proof dependencies and matrix unfoldings provide further specimens. Alphonic distinction is related to the provisional Greene–Pascal framing of the trajectories needed to sustain its functional use.

The general audit names a construction and its intended continuation, unfolds relevant trajectories, records licences and compressions, examines correspondences, and reports local sufficiency, witnessed insufficiency, representational debt and cost under stated conditions. Worked cases concern infinity, a finite set with two construction histories, a polynomial representation bridge, and the partition of a registered extent. The audit is itself a finite report whose language and stopping conditions remain inspectable. Function is retained throughout: mathematics, measurement, poetry and fiction can each sustain Functional Symbolic Trajectories under different commitments and regions of use.

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Notation and reading conventions

The abbreviation FST continues to mean *Functional Symbolic Trajectory*. Finite is a commitment concerning its actual instantiation; functional identifies its capacity to constrain and enable continuations. The following notation carries forward Parts I and II.

Sign	Working use
$R = (s, \alpha, \delta, H, C)$	A conditioned representational state: symbol, distinction condition, uncertainty, accessible provenance and licence.
Δ	A finite interactional or registration window.
Γ, F_H	A Functional Symbolic Trajectory; the second form makes accessible history explicit.
Λ_t	The present endogenous lattice acting as a handler.
L	An accessible finite ledger of relevant actions and distinctions.
$\mathcal{F}_C$	A finite report gathering trajectories and admitted connections for a purpose.
$\text{val}_C(L) = v$	Extraction of a value from a ledger under a stated reporting rule.
$T_{a,C}(L)$	A proposed ledger continuation, possibly unavailable under the stated conditions.
$\text{Adm}_C(T_{a,C}, L)$	An admissibility judgement concerning that continuation.
$A _C B$	Conditioned correspondence; no identity of carriers or histories is declared.
$\kappa_{\Lambda,C}, D_{\Lambda,C}$	Compression and handler-dependent reconstruction.
Φ, Ψ	Declared mappings between representations or trajectories.

The ordinary equals sign performs its declared job inside a formal domain or an explicitly stipulated value rule. The interactional bar marks conditioned comparison when registrations, histories or empirical application are at issue. This distinction does not require every printed equality to acquire an uncertainty interval.

Set-theoretic formulas use standard abbreviations. Their underlying language has membership as its non-logical relation and equality within first-order logic. Separation and Replacement are schemas: their displayed letters for formulas do not introduce quantification over all predicates inside one first-order axiom. Parameters are allowed and are universally closed as described in Appendix C.

Prologue: The Foundation at the Top

Imagine a mountain assembled through acts of measurement. Its base is broad enough to stand upon. We push, weigh, compare an extent, repeat a motion, mark a surface and recognise a difference. These acts already have instruments and language, but their relationships can be encountered through many overlapping trajectories. A registered whole can be turned in the hand. Another person can point to it. A balance can compare it with another. Their correspondences form a practical ground for further refinement.

The ascent requires more supporting work. An instrument narrows a window of distinction. A calibration relates its response to a reference. A correction gathers several records into a value. A mathematical construction holds relations that a familiar noun can no longer carry unaided. Each achievement makes a new distinction available, and each requires a trajectory through materials, records, conventions and symbolic operations. The upper region of the mountain is an accumulation of such work.

Eventually the language changes. A highly refined endpoint is called fundamental. It becomes the opening noun of another explanation, and the explanation proceeds downwards from it towards the measurements from which it was formed. The trajectory of its production and the trajectory of its explanatory use now run in different directions. This reversal can be highly functional. A stable compression allows a vast history to be carried by one term. The audit asks what happens when the second trajectory is presented without the first.

The same structure appears in mathematics. An axiom is placed at the head of a formal derivation. A group, number, function or space becomes an admitted object from which a proof can begin. The formal beginning is clear within the adopted rules. Yet the reader reached that beginning through a history of distinctions, words, training, inscriptions and agreement. The axiom is held within a theory of its use even when that theory is not written beside it.

Part I followed the middle between interaction and scientific statement. Part II followed the account between an action and its reported value. This volume follows the enabling trajectory behind a formal beginning. It asks how a construction becomes available, what it permits, what it compresses and how its next use is supported. The mountain is an image for this inspection; it is not a metric of increasing truth or a ranking of disciplines.

A finite audit cannot reconstruct every stone of the mountain. Some histories have become inaccessible. Some relationships need to be reconstructed through a present handler. Others can be checked in a surviving report. The relevant distinction is between declaring those conditions and allowing the apparent simplicity of the tip to remove them from the claim.

The inquiry begins with function. A person can build a useful mathematical path through an axiom, a poem through an imagined creature, or an instrument through a model. The different tellings do work. Their capacities are followed under the commitments that enable them. The task is to make those commitments available when the path is asked to continue into another operation or another basin.

CHAPTER 1

From Representation to Report to Audit

1.1 The ground carried from Part I

Part I begins with an instantiated representation. A mark occupies a carrier, becomes distinguishable across an interactional window, and enters a history of transformations. The visible symbol is therefore a selected position in a trajectory. Its scientific use depends upon the conditions under which it was formed and upon the correspondences under which it can return to registration [1].

The three basins organise this account. Physical interaction and registration supply changes across finite extents. The symbolic basin supplies operations on admitted inscriptions, values, diagrams and statements. The bridge basin holds the rules by which one kind of trajectory is related to another. These are working distinctions within a coupled process. An instrument can be designed symbolically and altered physically, while its output becomes a condition upon another symbolic trajectory.

The state

$$R = (s, \alpha, \delta, H, C) \quad (1.1)$$

is retained as a reminder of this architecture. Its components are not required to be stored in a single tuple inside every system. Accessible provenance can be distributed across several reports, and different steps can have different distinction conditions. What matters is that the audit identifies the components relevant to the present claim.

1.2 The account carried from Part II

Part II develops three paths through an operation: the bead path of admitted distinctions and actions, the ledger path recording their order and disposition, and the classical path operating on extracted values [2]. A finite report gathers one or more such trajectories into a coherent document. Its value can be detached and sent into a further calculation, but the document and value have different capacities.

The two-quarter specimen supplies an operational witness. Two ledgers yield the same proportion of a stipulated whole. One retains two unit beads; another retains one two-unit bead. Under the declared rule of selecting whole beads, the first supports another halving and the second does not. The arithmetic quarter remains available in both cases. Its sufficiency depends on what it is subsequently asked to decide.

This gives the audit a concrete beginning. A difference in histories alone does not establish insufficiency. The report names the next action or comparison for which the difference matters. Debt likewise names a particular missing distinction required by a claim. It does not become a substance attached to every short expression.

1.3 The new work of Part III

The present volume applies this machinery to inherited symbolic constructions. An axiom is inspected through its formal permission and the representational episode carrying its use. A mathematical object is inspected through its definition, operations and compressed dependencies. A bridge is inspected through the relationships it preserves and the continuations that need those relationships.

This extension does not require a fresh registration at every mathematical step. A formal proof can proceed through its admitted symbolic rules. The audit follows the rules, their instantiation and any proposed translation or application. The depth of unfolding is selected by the task. An imported lemma can remain a cited dependency when its conclusion is sufficient; another question may require the lemma's proof to be reopened.

The outcome is a further finite report. It carries findings that can themselves become portable beads. Its scope and stopping point therefore matter as much as those of the object being inspected. Representation, report and audit are successive functions within one continuing trajectory, not three completed containers.

CHAPTER 2

Two Starting Positions

2.1 The classical pointer

In this volume, ZFC sometimes serves as a practical pointer to the broad classical mathematical tradition whose modern foundations can be expressed set-theoretically. When a particular axiom is audited, ZFC denotes the declared formal theory. The two uses are marked because an algebraic practice, a classical logical rule and a set-existence axiom do different work. A lineage of mathematical use cannot be compressed into the assertion that every historical calculation began with the present axiom list.

The formal theory establishes a domain of objects and permitted inferences. The symbols of its language are interpreted through membership, equality and the adopted logic. The particular ink or memory carrying an occurrence is not made an argument of each membership relation. This abstraction enables the relation to be reconstructed across many carriers.

FSM places the conditions of representation at the beginning of its own account. Distinction, extent, uncertainty, provenance and admissibility belong to the construction through which an object becomes available. The order of inquiry therefore changes. Before asking which operations a named object permits, FSM asks what trajectory holds the name and makes those operations traversable.

2.2 Representation and commitment

A finite report can contain a classical definition of the real numbers, a statement about infinity, or a proof involving complex quantities. Their inscriptions occupy finite carriers and their reading unfolds through FSTs. That availability does not by itself make every commitment expressed by the classical definition a commitment of an FSM object.

There are consequently three inspectable positions: the finite inscription, the formal construction described by it, and the operative episode in which it is used. An infinite set can be licensed at the second position while its statement and present proof episode are finite at the first and third. The audit follows their relationship. It does not substitute the shortness of the inscription for the internal content of the definition.

For an FSM translation, a further act is required. The construction must be specified under the finite conditions adopted by FSM, and its relations to the classical construction must be declared. A finite approximation, a translated operator and an inspected classical inscription are different possible products. Their licences and capacities should remain available in the report.

2.3 The burden of carrying

The comparison can be followed through containment, distinction and continuation. A set presents membership under its formal rules. An FST carries ordered operations and available history. A formal equality permits substitution under the domain's rules. A registered correspondence requires the conditions under which two trajectories are admitted as corresponding.

Neither framework removes the need for a reader or machine to traverse an actual instance. The difference concerns where the instance and its conditions enter the account. FSM makes their role explicit and then examines the compression that permits formal mobility. This makes the classical construction a particularly rich specimen because its operations can be followed without requiring the report to adopt all of its foundational commitments.

A useful audit preserves the purpose of the comparison. A compact formal result may be sufficient for another formal proof. A new question about its production, algorithmic use or measurement application may require a longer trajectory. The report should state which of these functions is under inspection before drawing a conclusion about support.

Before the Axiom

3.1 A formal beginning with a history

An axiom begins a derivation under a declared formal licence. The operations following it are judged within that licence. This formal priority is a relationship in the adopted system. The act of writing, reading or admitting the axiom already has a trajectory that the formal ordering need not display.

Consider even a short statement involving membership. A reader distinguishes letters, recognises quantifiers, separates a bound occurrence from a free one, and follows a convention for equality. These capacities were acquired through earlier trajectories. A machine checker requires a grammar, an encoding and an implementation of its inference rules. The axiom becomes available through those conditions before it can function as a first premise.

The distinction can be held schematically:

$$\Gamma_{\text{enabling}} \longrightarrow A \longrightarrow \Gamma_{\text{derivation}}. \quad (3.1)$$

Here A is an axiom occurrence or an admitted statement in a report. The first arrow concerns its representational availability; the second concerns its licensed mathematical use. The diagram does not infer the axiom from every event in the enabling history. It records the two functions of the present bead.

3.2 What axiomatic placement compresses

Once the axiom is stabilised, its earlier support can recede from view. A printed list is then read as a collection of beginnings. The language of the list is inherited, the logic is assumed, and the institutional and pedagogical routes that made the statements ordinary can become inaccessible. This compression allows mathematics to proceed efficiently.

An audit can reopen the compression at several depths. It may need only the formal grammar and a source for the chosen statement. It may examine the permission's historical development or a reader's reconstruction of its meaning. It may inspect the reliability of an implementation carrying the derivation. These are distinct tasks. Their uncertainty is not automatically a new formal premise in the theorem being checked.

What changes is the report of the theorem's use. An exact derivation under stipulated rules and uncertainty about the particular episode of checking it can coexist. The report can identify a verified formal path, an unchecked dependency, a transcription problem or a gap in provenance. Restoring the conditions of the episode does not require blending these differences into one undifferentiated error bar.

3.3 The FSM axiom under the same inspection

The Axiom of Finite Representation is also held by a supporting trajectory. Its words inherit distinctions concerning a carrier, instantiation, extent, uncertainty and history. The volume cannot place it beyond audit simply because it supplies the audit's present licence.

Its use declares a starting commitment for an inquiry. The inquiry then follows the consequences of that commitment through specimens and worked constructions. Claims that extend beyond those constructions require their own trajectories. For example, a universal physical value for the Alphonic limit cannot be extracted solely from the statement that registrations possess operative distinction conditions.

The present axiom is therefore simultaneously a compressed outcome of earlier work and a permission for the work that follows. Maintaining both positions gives FSM continuity without turning its terms into conditionless beginnings. The audit is able to inspect its ground while continuing to stand upon it.

The Licence to Infinity

4.1 A mark that travels

The essay *The Licence and the Line* supplies a historical and interpretative trajectory into this inquiry [4]. It follows marks, mathematical practices and institutions through which permissions become ordinary. Its relevance here concerns the travel of a licence: a symbolic expression can retain its spelling while its later uses compress the circumstances of its formation.

The finite mark ∞ can be registered, copied, spoken about and incorporated into a proof. It has a carrier and an occurrence history. Those facts permit FSM to inspect it as a symbolic bead. The mathematical commitments associated with a particular occurrence must then be recovered from its context rather than inferred from the shape alone.

An unbounded continuation instruction, an infinite-set assertion and a limit expression need different accounts. The same family of words can coordinate them, but their formal permissions differ. The audit begins by identifying which use is being inspected. This keeps a historical narrative from becoming an automatic equivalence between every mathematical appearance of infinity.

4.2 Continuation and completed commitment

A rule can permit a next step whenever a present state meets a condition. A finite episode applies that rule some number of times and records the results. A classical statement may additionally assert an object closed under the rule. The episode and the object asserted are not identical descriptions of the same thing.

For example, one can write a finite successor trace and also write the axiom asserting an inductive set. The trace makes specific steps available; the axiom supplies a formal permission concerning an entire set. Repeating the trace does not by itself enact the completed commitment. Conversely, describing the commitment finitely does not alter its classical meaning into a trace of only the displayed steps.

FSM can follow all these inscriptions. Its native construction must identify the finite continuation rule, the present admitted states and the conditions under which a further state becomes available. A relation to the classical construction can be declared. The report then identifies what that relation preserves for the task.

4.3 Neighbouring specimens

Zero requires its own inspection. A numeral, an additive identity, a reference setting, an empty ledger entry and an absence of detected response can occupy related language paths without becoming interchangeable. A detector code labelled zero still has a carrier and a registration window. An empty collection is a formal object under a particular licence.

The continuum likewise requires its definition and its use to be identified. A mathematical construction can license exact internal operations, while a finite array or registered extent can supply only the distinctions its report records. The audit follows the correspondence rather than allowing a shared noun such as line to settle the relation.

The historical work is carried as provenance and as an interpretation that can be further unfolded. This volume uses it to make permission visible. Detailed historical attributions require their own source records; they are not derived from the finite successor or bead specimens. Those specimens perform a narrower and inspectable function within the present mathematical audit.

The Container and the Trajectory

5.1 Membership as organisation

A set presents its members together under a membership rule. In the chosen ZFC setting, formal objects are sets, and other constructions can be implemented through them. The organising relation is exceptionally portable: two constructions can be treated as the same set when they have the same members, regardless of the order in which a particular presentation was assembled.

Order and history can themselves be encoded set-theoretically. The comparison here concerns what the chosen presentation carries for its task. An unordered membership presentation need not display a ledger simply because the formal system has resources with which a ledger could be encoded. The availability of a possible encoding and the presence of an accessible account are different positions.

FSM locates the carrying structure in the FST. A set definition, its assembly, reading and use can all occur within such a trajectory. The set becomes an admitted compression that makes membership relationships available at a stage of that process. Its subsequent operation can be followed through the report's accessible connections.

5.2 Different histories at one presentation

Suppose two ledgers produce the same finite collection of labels. Their membership reports agree. One ledger may retain the source and order of each entry; another may carry only the present labels. If the proposed continuation is a membership test, the compact report may be sufficient for both. If the continuation asks which entry was admitted first or under a particular calibration, another record is needed.

This is a change of question, not an intrinsic defect in membership. The report must expose the transition from asking what belongs to asking how it became present. The second question needs an FST or a suitably encoded account that was not part of the first presentation.

The same reasoning applies to a compound report. It gathers several FSTs and their admitted relations. A table of final values is one compression of that document. A dependency record is another. Their functions can overlap while their capacities for continuation differ.

5.3 The wrapper as a continuing act

The term wrapper points to what holds the construction together for its present use. In a set presentation it is a membership organisation. In an FST account it is the ordered, history-conditioned continuation of relations. The latter can carry the former and can also carry the act of changing from one presentation to another.

There is no requirement here to create a completed set of all FSTs as the container of the audit. The Grand Corpus points to the wider accumulated and continuing field of symbolic trajectories through which a local expression is held. A particular report accesses only a bounded portion of that support. Its accessible references and present handler determine what can be reconstructed.

The wrapper therefore remains dynamic. New registrations, definitions or readings can alter the available continuations. A finite report closes with respect to a declared task, then becomes a bead in another trajectory. The audit follows where that closure was made and what a later use asks it to carry.

An FSM Audit of the ZFC Axioms

6.1 The presentation and the audit

The specimens in this chapter are stated in classical first-order logic with equality and the membership relation. All variables range over sets. A formula schema is specified externally by its pattern, with a separate instance for each appropriate first-order formula. The presentation is deliberately redundant: the empty-set statement is shown separately, and Separation is retained for inspection alongside Replacement. The audit is concerned with the distinct jobs these permissions perform, rather than with finding the shortest equivalent axiom list [5, 6].

A formal formula declares its own quantifier scope. The audit adds an account of a particular use: its written occurrence, cited dependencies, transformations and purpose. It then asks what a proposed FSM construction would need to carry. These levels should remain explicit. An existence assertion is not an instruction to enumerate every member, and the audit does not convert it into such an instruction merely in order to inspect its cost.

Each specimen below therefore has a formal statement, an operational reading and an audit focus. The paragraph describing the audit is a construction of the present theory. The classical statement supplies the permission being inspected. A finite procedure can exemplify that permission locally, while a fuller classical commitment may remain part of the formal licence.

6.2 Extensionality

A convenient statement is

$$\forall A \forall B [\forall x (x \in A \leftrightarrow x \in B) \longrightarrow A = B]. \quad (6.1)$$

The permission concerns identity through membership. A finite example with the same two members presented in different orders illustrates how the present set is held independently of the order of inscription. Equality then makes substitutions available within the formal domain.

The FSM audit asks which differences the membership presentation leaves outside that identity. A ledger can record that one member was admitted before another, or that two presentations were produced under different registrations. Those records do not become additional members simply by having occurred. They can be retained in a separate account or encoded in a richer construction.

The profile therefore distinguishes the equality of formal sets from the correspondence of their representational episodes. The equality can support a membership-based continuation. A claim concerning the order of admission requires another trajectory. Extensionality does not supply that trajectory, and its absence from the axiom does not establish that no suitable record can be made.

6.3 The empty set

The existence statement

$$\exists E \forall x (x \notin E) \quad (6.2)$$

makes an empty collection available. Extensionality gives uniqueness. In the redundant presentation used here, the statement is listed explicitly; its existence can also be recovered from the other permissions. For example, given a set A , a Separation instance with the contradictory condition $x \neq x$ gives a subset with no members.

The inscription $\emptyset$ is itself a distinction. A ledger reporting no admitted entry also occupies a carrier. An empty collection, a report of no detected response and a missing file therefore have different histories and functions. A reader should not infer one from another solely because the same word empty appears.

The audit profile names the role of emptiness in the task. An empty formal set can initialise a recursive construction. A detector record can state that no event passed a threshold in a chosen window. A missing record can state only that an account is presently unavailable. An FSM translation retains the registration and licence needed to distinguish these uses.

6.4 Pairing

The statement

$$\forall a \forall b \exists P \forall x [x \in P \leftrightarrow (x = a \vee x = b)] \quad (6.3)$$

licenses the unordered pair $\{a, b\}$. When $a = b$, the same permission gives a singleton. A written example can be unfolded through the admission of two labels and the formation of a membership presentation.

The audit asks how the entries are recognised and what relationships the pair is being asked to preserve. The two labels may be sufficient for an internal membership operation. If the pair stands for two registered specimens, their individual histories and the map from specimen to label must be declared. Writing two names together does not itself register their physical relation.

Pairing also prepares more elaborate encodings, including an ordered-pair construction through nested sets. The order then belongs to the encoding convention and its proven properties. The FSM report can follow those steps instead of treating order as silently supplied by the visual left-to-right placement of braces and commas. The formal construction and its inscriptional layout can cooperate without being collapsed.

6.5 Union

The Union permission can be written

$$\forall A \exists U \forall x [x \in U \leftrightarrow \exists B (B \in A \wedge x \in B)]. \quad (6.4)$$

It makes the members of members available in one collection. In a finite example, two admitted groupings can be followed through this reorganisation and their union subsequently used as a single membership presentation.

The audit records whether the earlier grouping remains accessible. If a later task asks which source group contained a particular member, the union alone may not decide it. That history

can be retained in the source family, a relation table or a ledger. The operation's functional gain is consolidation; its sufficiency depends on the task given to the consolidated presentation.

No destruction of a source record follows from Union as a formal assertion. The audit concerns the particular compression carried forward. It distinguishes a report that cites the original family from one that propagates only the resulting members. The relevant debt arises when the latter is asked a question requiring the former.

6.6 Power Set

The statement

$$\forall A \exists P \forall B [B \in P \leftrightarrow B \subseteq A] \quad (6.5)$$

uses $B \subseteq A$ as the abbreviation for $\forall x (x \in B \rightarrow x \in A)$. The formal permission concerns a collection of subsets. A small finite case can display the entire collection under its stated rules.

For $A = \{a, b\}$ with distinct members, the local construction gives

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}. \quad (6.6)$$

The audit can follow a two-position inclusion code and its mapping to these four subsets. It can count the required operations in this chosen procedure. The compact symbol $\mathcal{P}(A)$ and the displayed enumeration have different inscription costs while preserving the relevant membership relation.

The general classical permission is not restricted to this displayed case. An FSM construction must identify which subsets are actually admitted and how the intended use is supported. A finite all-subsets enumeration, a compact rule, and an existence claim concerning an infinite source set have different profiles. The audit records which one is present and which continuation the report requires of it [7].

6.7 Infinity

Using the previously licensed abbreviations, write

$$\exists I [\emptyset \in I \wedge \forall x (x \in I \rightarrow x \cup \{x\} \in I)]. \quad (6.7)$$

This asserts an inductive set. The set supplied by the assertion is not automatically the least inductive set; that further construction and its designation require additional steps. This matters because the name attached to an axiom's witness should not silently carry a later definition.

A finite successor trace can exemplify particular transitions. The written record admits the displayed stages and the rule used to obtain them. The classical permission additionally supports its infinite-set commitment. The audit follows the distinction rather than treating a finite initial trace as either an enactment or a refutation of the completed claim.

The FSM profile asks what present trace, continuation rule and supporting report are available. An unbounded licence to continue is still expressed through finite episodes of use. A claim about a whole formal domain can be followed under its classical licence; a native FSM object requires its own conditions. Chapter 18 develops this specimen.

6.8 Separation

For each suitable formula $\varphi(x, \vec{p})$, with B not free in that formula, a schema instance is

$$\forall \vec{p} \forall A \exists B \forall x [x \in B \leftrightarrow (x \in A \wedge \varphi(x, \vec{p}))]. \quad (6.8)$$

The source collection A remains part of the statement. The displayed φ is a pattern for formulas, not a new predicate variable quantified within a single first-order sentence.

For a finite list, a declared predicate can be evaluated along a particular traversal. The ledger can retain accepted entries, rejected entries and the order of testing. If the compact subset is used later, the test record may become relevant to a question concerning selection or changing criteria.

The classical schema is an existence permission, not a guarantee that every predicate has an available decision algorithm. An FSM selection procedure consequently records the predicate, source, admitted test operation and treatment of unresolved cases. A complete formal subset and a partial finite classification report must be related by a declared correspondence.

6.9 Replacement

For each appropriate formula $\theta(x, y, \vec{p})$, with B not free, use

$$\begin{aligned} \forall \vec{p} \forall A [\forall x \in A \exists! y \theta(x, y, \vec{p}) \\ \longrightarrow \exists B \forall y (y \in B \leftrightarrow \exists x \in A \theta(x, y, \vec{p}))]. \end{aligned} \quad (6.9)$$

Its displayed functional condition is stated on the source set. Different source members may still have the same image. Uniqueness of output per source does not declare an injective mapping.

A finite operation such as mapping two labels to one category makes this point visible. The image presentation has one category, while a source-to-image ledger can retain both routes. Reconstructing a unique source from the image would require another permission and another record.

The audit profile identifies the formula, source domain, function condition, image and accessible mapping episode. If a later operation needs the preimage history, the image alone may be insufficient. If it needs only a property established for every image member, the compact result may suffice. Replacement and its operational realisation are followed at those different levels.

6.10 Foundation

A common statement is

$$\forall A [A \neq \emptyset \rightarrow \exists b (b \in A \wedge b \cap A = \emptyset)]. \quad (6.10)$$

The intersection abbreviation records that no member of b is also a member of A . The permission constrains membership structure.

A temporal return to an earlier symbolic state is a different relationship. A reader can revisit a paragraph and a calculation can revisit a stored expression without making a set a member of itself. The audit must identify the relation under inspection before borrowing the language of cycles from a diagram.

The FSM profile therefore asks which membership constraints are adopted and which constraints govern its own continuation graph. A feedback trajectory, a reference between documents and set membership have separately declared roles. A correspondence between them can be constructed, but it is not established by calling all three recursive.

6.11 Choice

Use the choice-function form. For each set A whose members are non-empty sets, there is a function f with domain A such that $f(B) \in B$ for each $B \in A$. Written with the function and domain abbreviations, the statement is

$$\begin{aligned} \forall A [\forall B \in A (B \neq \emptyset) \\ \rightarrow \exists f (\text{dom } f = A \wedge \forall B \in A f(B) \in B)]. \end{aligned} \quad (6.11)$$

The assertion does not attach one selection procedure to every instance.

For a finite displayed family, one can record particular selections under a rule. The audit then identifies the source entries, selection operation and retained choice record. Such a finite demonstration does not identify the full general permission with that algorithm.

An FSM continuation requiring an actual selection needs an available selecting trajectory or an explicitly stated unresolved obligation. A formal proof requiring only the existence conclusion has another profile. The difference between existence, exhibited selection and performed selection should remain available when the result travels.

6.12 The profiles gathered

The chapter has followed a common architecture across different permissions. Identity, collection, reorganisation, selection, mapping, closure and structural restriction each support different symbolic functions. Their finite examples make those functions inspectable without pretending that one example exhausts the corresponding formal assertion.

The audit profile records the formal domain, finite episode, dependencies, compressed distinctions and proposed continuation. It can state that an internal formal move is admitted while an FSM translation remains to be constructed. It can also state that a particular finite implementation already supports a bounded use. Neither statement requires a verdict about every possible use of the axiom.

This comparative record now becomes a bead for the chapters on objects, algebra and bridges. Its details remain accessible when those chapters need to reopen a permission. The conclusion supplied here is a finite map of the selected architecture and its inspection questions.

CHAPTER 7

The Classical Mathematical Object

7.1 A name, a construction and a use

The noun object permits mathematics to gather relationships into a portable unit. A number, function, matrix or group can then become a starting point for further operations. The name is economical because the reader's training and the report's references reconstruct enough of its supporting trajectory.

The audit separates the name's inscription from the definition it invokes and the present operation using it. A short word can lead into a long construction; a compact definition can support several different algorithms. The finite occurrence is not an inventory of every relationship licensed by the definition.

An object can consequently be inspected at different resolutions. A theorem may need only its domain and an established property. An implementation may need an encoding, access order and arithmetic regime. A translation into FSM may need a construction that carries distinction, extent, uncertainty and history. The name alone does not settle which report is sufficient.

7.2 Zero, natural numbers and real numbers

Zero provides an elementary specimen. Its formal function can be specified within an arithmetic structure. Its occurrence in a measured report can instead name an admitted response region, a reference origin or the absence of a counted event. Those uses can be connected, but the bridge must say what is preserved.

A set-theoretic construction of natural numbers can be followed through finite displayed stages and formal closure permissions. A counting episode can separately follow a registered collection. A numeral shared by the two reports allows a comparison under a counting rule; it does not identify the whole set-theoretic construction with the counted carrier.

A real number likewise has an internal classical definition. A measured value enters through a finite registration and its conditions. A finite decimal, a formula specifying a classical real, and a detector code can each be short inscriptions with different continuation licences. An audit should state which is being used before interpreting precision or uncertainty. More digits in an inscription need not provide another registered distinction.

7.3 Complex quantities and functions

A classical complex quantity can be carried through its algebra and through a paired real-coordinate representation. The written i can participate in productive formal trajectories. Its finite inscription does not make an independently measured imaginary quantity available as a native FSM object.

The FSM programme asks how measured or finitely admitted components and their relationships could support an alternative object. Delay coordinates and embedded trajectories are possible directions for that construction. A claimed replacement would need its operators, domain, return operations and correspondence with the classical use to be specified. Calling a trajectory embedded does not complete that work.

A function supplies another distinction between permission and procedure. Its formal graph or defining relation specifies an internal relationship. An algorithm computes particular outputs under stated rules. An instrument realises a physical trajectory whose output can be compared with those outputs. The audit identifies each layer and the task for which its present account suffices.

7.4 Matrices, integrals and groups

A matrix organises entries for operations. Its printed arrangement makes some relations quickly readable while its actual use unfolds through access, stored state and transformations. Chapter 13 makes this operational distinction explicit.

An integral can be inspected through its chosen definition, the conditions of a theorem involving it, a finite quadrature procedure, and a measurement comparison. These are connected constructions. A summation report does not become an exact classical integral solely through a change of label, nor does the presence of an ideal integral remove the need for finite steps in a particular computation.

A group can be inspected through its operation, identity and inverse permissions. A finite operation table is an especially accessible specimen because its entries can be followed and its rules checked under the declared domain. A physical action associated with an entry requires another trajectory. The audit records which representation and operation are being asked to travel.

The present volume leaves fuller object reconstruction open. Its contribution is to identify the account a proposed reconstruction would need. A finite object is then introduced through its constructive trajectory and permitted continuations, rather than acquired solely by attaching the adjective finite to a classical noun.

Algebra as Symbolic Mobility

8.1 The licence to move

Algebra makes constructions mobile. An expression can be transformed through substitutions and cancellations, and a result can enter a new proof without carrying every previous line beside it. The compact expression is a handle held through a grammar and a licence.

The familiar relation

$$(a + b) - b = a \tag{8.1}$$

performs an exact job in an appropriate additive structure. Its report need not name a particular gold transfer or detector response. Those belong to another use of the expression. The sign's algebraic function is preserved by its domain conditions.

In an account of a transfer, the retained amount, displaced amount, action and accessible disposition record can all be relevant. The same value result may be obtained through several such ledgers. Algebra supplies mobility for the values; the ledger supplies support for questions concerning their particular formation.

8.2 Implementation as another trajectory

A numerical implementation adds an encoding and a sequence of operations. A formal identity does not by itself determine every result of a finite arithmetic implementation. As a stipulated decimal example, take two significant decimal digits after each operation. Adding 0.4 to 10, rounding the intermediate result back to 10, and then subtracting 0.4 gives 9.6, rather than the formal result 10.

The specific outcome depends on the declared rounding and tie rules. The audit therefore records those rules rather than attributing the difference to algebra in general. Parenthesisation, storage and the stage of rounding belong to the operation being performed.

This implementation account is itself symbolic. It can be described mathematically and checked through a finite trace. The point is to retain the licence under which each manipulation occurs. A formal substitution and a device operation may share an inscription while preserving different relationships.

8.3 What portability costs and enables

Compression reduces the immediate burden of reading and carrying a result. It can preserve exactly the relationship a further proof needs. Reopening every dependency at every use would create another cost and could obstruct the task it was meant to support.

The audit consequently asks for an appropriate depth of unfolding. A source reference may suffice for one dependency; an independently checked transformation may be needed for

another. A named missing condition opens a debt entry when it matters to a specific use. The brevity of the expression alone does not establish that debt.

Algebra's function is therefore recorded positively. It enables substitution, reuse and construction across many paths. FSM follows the conditions under which this mobility can be used and the points at which a later question needs the supporting account to become visible.

Perfect Division and Finite Distinction

9.1 The admitted boundary

A formal system can license exact distinctions. Membership, equality or a truth convention can separate cases under its declared rules. A finite registration instead supplies the distinctions that its operation makes available. The two accounts can be related without giving the same role to each boundary.

The Alphonic condition identifies where a relevant difference becomes non-single in the present operation. It is specified with the apparatus or observer, carrier and licence. The present formulation does not require one numerical threshold to apply to every stage of a compound report.

Two candidate states can be admitted as the same code by one operation and separated by another. The latter is a new distinction event. It should enter the ledger with its conditions. The earlier report cannot acquire it merely because a later algebraic expression writes two distinct labels.

9.2 Uncertainty and formal identity

Uncertainty belongs to a registration or claim under finite conditions. Its form can change during transformation and comparison. Formal identity belongs to a declared symbolic domain. An exact internal equality and uncertainty about how a particular occurrence was registered can therefore be carried together.

This distinction matters when a proof is read. A reader can follow an exact formal step while remaining uncertain about a transcription, omitted premise or interpretation of a symbol. Those uncertainties concern the present reconstruction. The audit locates them instead of spreading one uniform uncertainty value over every formal relation.

A measured comparison requires its own criterion. Matching visible numerals can be useful for a decision, but the histories and uncertainty structures may still matter to another task. The interactional bar records that the comparison has stated conditions. It does not diminish a precise comparison; it identifies what makes the precision available.

9.3 Device logic as a specimen

A device can admit response regions as codes labelled zero and one. Its report can record threshold conditions, an observation window and the treatment of responses not securely admitted to either region. The symbolic labels then enter further operations.

The formal logic used to design or describe the device and the device's physical distinction process have different jobs. The mapping between response and code belongs to the generonic chain. Its successful use permits compact symbolic operations downstream.

This specimen shows how finite distinction supports powerful exact symbolic conventions in a local regime. The audit can follow their relationship without declaring the code to contain the entire event. A later change of threshold or window becomes another act whose effect can be compared through an extended report.

Greene–Pascal: Representational Support

10.1 The provisional framing

The Greene–Pascal contribution is introduced here as a framing, open to further construction: the FST, or relevant region within the Grand Corpus of FSTs, needed to hold and functionally use an Alphonic distinction. The starting question concerns representational support. What permits the distinction to remain available through naming, reading, reconstruction and subsequent operation?

A registered difference is not held by a solitary glyph independently of its handler. Its code requires a convention. Its measurement use can require a reference, calibration and uncertainty account. Its linguistic use can require other words and prior trajectories. A local symbol is therefore supported by relationships extending beyond its visible carrier.

The Grand Corpus points to this wider field of accumulated and continuing symbolic relationships. A finite audit accesses a bounded region through its records and present handler. It does not enumerate a completed totality of every possible FST. The region of support is reconstructed for a specified use.

10.2 What it means to hold a distinction

Holding can be followed through several activities. A system must distinguish the relevant mark from neighbouring states. It must retain or reconstruct the relationship that the mark is licensed to carry. It must place that relationship into an operation. If the construction travels to another reader or system, enough support must be available for a corresponding reconstruction.

Consider a code admitted by an instrument. In one task the code and unit convention may support a simple threshold decision. In another, comparison across instruments requires calibration history and uncertainty. The same code now calls upon a larger supporting region. The difference is produced by the named continuation.

The framing therefore does not yet specify one shortest chain in every context. A short inscription can depend on a highly trained handler. A longer report may make the same relation available to a less specialised reader. The visible chain length and the supporting trajectory burden need to be distinguished before a minimum can be proposed.

10.3 Support, refinement and cost

Refinement can alter the support needed to hold a distinction. A new apparatus may require a more elaborate calibration chain. A different encoding may shorten the portable record while increasing the construction needed to interpret it. A revised report may make an earlier relationship usable with less reconstruction.

These possibilities prevent a single monotonic law from being assumed here. More refined distinctions can require more work in a particular trajectory, but the comparison must say which carrier, procedure, handler and function are held fixed. A better compression or new instrument can change the relationship.

Cost can then be recorded in appropriate quantities: operations performed, stored characters or bytes, reading time, instrument use, or other declared resources. These entries should not be added into one scalar without a conversion rule. Representational debt remains the missing distinction owed by a claim; cost records the work of carrying or recovering support.

10.4 Questions for development

One development question asks for a minimum sufficient accessible account for a given continuation. Another asks how that account changes across handlers or resolutions. A third asks which supporting relationships can be compressed while the named operation remains reconstructable.

A bounded comparison could exhibit two reports that support the same operation under a declared test, then record their different costs. An insufficiency witness could show where further compression removes a needed relationship. These specimens would give Greene–Pascal a constructive route beyond its present framing.

The contribution remains tied to Alphonic distinction through this support relation. It is not defined here solely as a resource threshold at which inquiry must stop. Questions concerning functional gain, resource expenditure and continuance can follow from the work of sustaining the distinction, with their own declared measurements and licences.

Bridges Between Mathematical Fields

11.1 A correspondence with a purpose

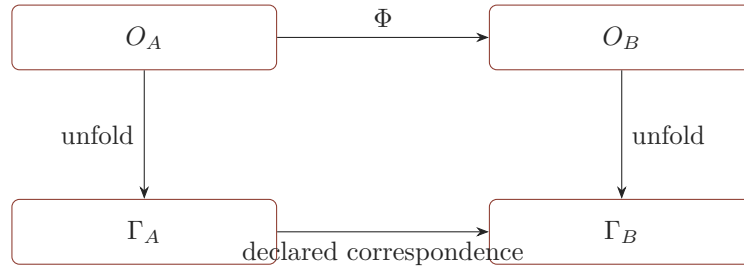
A mathematical bridge permits a relationship established in one representation or field to support work in another. The mapping may preserve an operation, an ordering, a value or a structural property. Its function depends on that preservation being sufficient for the proposed continuation.

The audit begins by naming the source, target and mapping. It also identifies the task for which the mapping is being used. A label such as equivalent or isomorphic can be unpacked through its domain and preserved relations. Those relations are the licence under which a result travels.

A present use then unfolds through finite operations. The source construction is read, translated, manipulated and perhaps returned to a source representation. The report can retain intermediate states or compress them into a theorem reference. Both forms can be functional, but they expose different portions of the route.

11.2 Unfolding both sides

Let O_A and O_B be two compressed presentations. Their relevant operational unfoldings are Γ_A and Γ_B . The audit inspects the bridge at both levels:



The lower correspondence is not inferred solely from the visual diagram. The report specifies which relations in the unfoldings need to correspond. A terminal value map can be sufficient even when the two procedures have different intermediate states. Another task can require access to those states.

Suppose a source operation T_A is proposed and a target operation T_B is used to perform corresponding work. The required relation may be represented schematically by

$$\Phi(T_A(\Gamma_A)) \mid_C T_B(\Phi(\Gamma_A)). \quad (11.1)$$

The expression is a reading aid. The domain of Φ , the operations and the comparison have to be declared. In an exact formal setting, a corresponding algebraic equation can replace the bar under its licence. In a registration comparison, the bar retains the finite conditions.

11.3 Direction and return

A bridge need not be reversible. An image can gather several source constructions into one target. Even where an inverse exists formally, a particular finite implementation or compressed report may require additional information to perform the return.

The audit therefore separates a forward map, a reconstruction map and an inverse with proven domain conditions. It asks whether the intended result needs recovery of a value, a canonical presentation, an original carrier or a particular history. Those are different return tasks.

Mathematical bridges also remain distinct from empirical crossings. A map between two formal presentations can be checked internally. Relating either presentation to a registered extent introduces another correspondence. The three-basin architecture preserves this distinction while allowing the report to follow the full route.

11.4 Parallel trajectories

The language of parallel FSTs points to relationships that remain usable across different tellings. One path may employ algebra and another a diagram. Their relevant nodes and continuations can correspond without their entire histories becoming identical.

Part III can use this comparison to inspect bridge-dependent proofs. A result reached through one field becomes a condition for a continuation in another. The audit identifies which preserved relationship licences that transition. This makes feedforward through the wider symbolic lattice available as a constructive account of mathematical development.

The supporting report can then become a new portable bead. Its function is to retain the conditions needed for later reuse. A proposed bridge that preserves less than the next task requires opens a particular debt; a bridge that preserves enough supplies a local sufficiency finding.

Proof as a Compound Functional Symbolic Trajectory

12.1 The report carrying the proof

A proof contains more than its concluding statement. Definitions establish available expressions, lemmas compress earlier routes, substitutions move relations, and cited theorems permit steps that would otherwise require longer arguments. Its reader follows these connections through a handler formed by prior training.

A finite proof report can therefore be inspected as a compound FST. Its printed order supplies one reading route; its dependencies supply other routes. Reopening a lemma can lead through another report and then return to the present argument. The audit chooses the needed depth and records its stopping point.

The theorem is a portable conclusion under the proof's licence. A later proof can use it without requiring the original author or the original paper carrier. A question about which algorithm or construction produced the result may need the wider document. The audit distinguishes these uses before deciding what should be unfolded.

12.2 Dependency and transition

A transition can be written as

$$s_i \xrightarrow{r_i, C_i} s_{i+1}, \quad (12.1)$$

where r_i records the invoked rule, definition or result and C_i its relevant conditions. This does not require every rule to be written beside every line in the original proof. It is a finite reconstruction selected for inspection.

A dependency record can identify the premise, its source and the status of its present checking. An audit may follow a quoted lemma's exact statement without unfolding its proof, provided that this depth is sufficient for the named task. If the use depends on an omitted condition, the record can expose the point at which another report is needed.

A bridge transition adds its source and target domains and the relationship being preserved. Returning to the original proof then requires the conclusion to meet the conditions of the next rule. The audit follows the whole excursion, including the return. A visually simple implication arrow may thus gather several distinct trajectories.

12.3 Proof dynamics and possible embedding

Different proofs of a result can preserve different intermediate constructions and make different later problems accessible. The same final theorem can therefore be carried by reports with different continuation capacities. This is a functional distinction that can be investigated without assigning one path the entire meaning of the result.

A later dynamical analysis must specify how the proof is unfolded. States could be admitted expressions, rule applications, dependency positions or parameterised construction records. An embedding procedure would then have a declared observation sequence, coordinates and comparison purpose.

This volume does not infer that a plotted trajectory is automatically a complete invariant of a proof. Its contribution is to make the necessary choices explicit. The attractor, reachability or stability questions belong to the chosen reconstruction and need their own report. The finite audit prepares that report by exposing the operations that a compressed proof carries.

Order Is Not Dimension

13.1 The ground of the unfolding

Within the FSM account developed here, the carrying lattice has finite extent in the ground three-dimensional setting. Higher order indexing organises relationships that are realised through inscriptions, stored states and time-extended operations. The index order does not itself add measured spatial directions to the carrier.

A matrix can consequently be a compact presentation of many relations while its actual operation unfolds sequentially. Its entries need an encoding and an order of access. An intermediate state must persist sufficiently for the next operation to use it. The written rectangular arrangement helps a reader reconstruct those relationships.

An unfolding is selected for a purpose. It can preserve the entries and the intended operation while making a particular access order visible. The finite report records that choice. It does not infer that every relational feature of every abstract coordinate construction is preserved by an arbitrary three-coordinate picture.

13.2 A matrix operation in time

Take a formal matrix and column vector,

$$M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad u = \begin{pmatrix} 5 \\ 6 \end{pmatrix}. \quad (13.1)$$

One admitted row-wise trace is

$$\begin{aligned} r_0 &= 0, & r_1 &= r_0 + 1 \cdot 5 = 5, & r_2 &= r_1 + 2 \cdot 6 = 17, \\ q_0 &= 0, & q_1 &= q_0 + 3 \cdot 5 = 15, & q_2 &= q_1 + 4 \cdot 6 = 39. \end{aligned} \quad (13.2)$$

The terminal report is $Mu = (17, 39)^T$. The trace displays the source access and intermediate sums. Another algorithm could traverse the entries differently and reach the same formal result.

The formal arithmetic here is exact under its stated domain. A finite implementation must add its arithmetic and storage conventions. A physical application must add the registrations and comparison rules that relate the entries and outputs to measured quantities. The matrix noun can gather these routes, but the audit keeps their licences available.

13.3 Indexing, carrier and continuation

A tensor adds further indexing conventions. Its use still requires finite acts of accessing and combining entries in the present episode. The number of indices and the carrier's measured extent answer different questions. An audit specifies both when both matter.

The same distinction applies to phase-space coordinates. A coordinate organisation can be constructed as a relational representation and then carried on a page or in memory. Its functional use depends on the chosen observation, transformation and comparison. The account of that use unfolds through finite states with time extent.

This chapter supplies a geometrical discipline for the subsequent object programme. Index order, representational organisation and measured spatial extent remain explicitly related through construction. Their shared use of the word dimension should not remove the trajectories that make each one available.

The Functional in FST

14.1 What a trajectory enables

A symbol functions by altering what can happen next. It can permit a calculation, constrain an imagined scene, trigger an instrument instruction or coordinate another person's action. Its effect is carried through a trajectory and a region of use.

The functional therefore cannot be reserved for a final declaration that a claim is physically measurable. Poetry, fiction and nonsense can produce and redirect trajectories. A mathematical construction can organise a proof. A detector report can support an engineering decision. Their relationships to registration differ, and those differences belong to their respective licences.

A useful audit starts by naming the function being sought. A fictional creature is not insufficient because a reader cannot weigh it in a laboratory. It may be sufficient for the narrative's next sentence. If that sentence is then used as a zoological report, another task and another bridge have been proposed.

14.2 The gold bar and its tellings

A gold bar can be held as a mass, an asset, a reference, a chemical specimen, an heirloom or a danger to a foot beneath it. Each telling selects and carries relationships that enable particular actions. Their shared pointing does not make the tellings interchangeable for every continuation.

An account of custody needs provenance and disposition. A weighing report needs its apparatus, reference and uncertainty. A monetary valuation needs another licence. The same visible numeral may occur in more than one account, but its function is held by different supporting trajectories.

The audit can follow these relationships without requiring one symbolic telling to exhaust the bar. A claim concerning a fracture or a transfer needs the appropriate registration route. A poem about the bar can function through metaphor and inherited language. The comparison is inspectable through what each telling does.

14.3 Four distinguishable questions

Internal coherence asks how a construction holds together under its rules. Functional adequacy asks whether it supports the task assigned to it. Exogenous tethering asks which registrations constrain it. Measurement admissibility asks which conditions license the stated empirical comparison.

These questions can interact. A scientific report may require a coherent mathematical argument, a functionally adequate instrument procedure and an admissible measurement

bridge. A mathematical proof may require the first two in a different form. The audit should identify which questions the present task calls upon.

This also prevents function from becoming an absolute score. A construction can be adequate for one operation and require another account for a neighbouring one. Function is followed locally, with its purpose and continuation. The ability of FSTs to enable different tellings is a resource for the inquiry rather than an obstacle to it.

14.4 The audit's own function

The audit asks a reader to spend resources unfolding a construction. That expenditure should make a relevant relationship available: a permission, dependency, omitted distinction or continuation that the compact presentation did not expose.

An audit that supplies no relevant distinction may need to be shortened or given another purpose. Its functional adequacy is therefore part of the self-audit. A longer report is not automatically a better account. Its value depends on what its added trajectory enables.

The present method preserves the ability to use short symbols. It supplies a route for reopening them when the task requires it. The report can then be recompressed with the necessary conditions attached or accessible through a reference.

From Fundamental to Functional

15.1 The endpoint reused

The mountain image can now be followed more precisely. A registered difference becomes a coded value, a transformed relation, a named object and a component of a theory. The object is then used to organise new measurements and explanations. It becomes a starting bead of another FST.

The audit asks which part of the producing trajectory is still available when that reuse occurs. A theory can be a highly functional organiser even though its production involved a longer instrumental and symbolic route. The function of organising does not recover every earlier registration by itself.

The distinction matters whenever the language of fundamentals makes the producing route disappear. The supporting account can be recovered to the depth needed by the new claim. Missing provenance can be named. A fresh experiment can supply another route. These acts extend the mountain rather than merely redescribe its tip.

15.2 Refinement and continuance

Refinement uses resources within a system that must continue. Instrument time, materials, computation, attention and institutional organisation can be recorded in an account. Their quantities have different units and their significance depends on the task.

The Greene–Pascal framing asks what support holds the new distinction. A further functional question asks what that support enables relative to its demands. The two questions are connected, but a cost comparison is another construction with its own measurement and licence.

Continuance can therefore remain an exploratory organising idea. A trajectory that sustains a system or community can acquire a persistent function. The report need not turn this observation into a universal scalar purpose. It can follow the particular relationship between the support of inquiry and the continuations that inquiry makes available.

15.3 Stabilising commitments

The conversation's questions about purpose extend this inquiry towards commitments that stabilise shared orientations or stopping points. Their historical and psychological trajectories remain available for a wider audit.

A mathematical licence similarly stabilises a field of work and ways of judging its continuations. The audit follows that function through its supporting history.

Part III carries these links as questions for broader audits. Here its task remains to follow how a refined endpoint is produced, held and reused.

A General Geofinite Audit

16.1 Provenance and purpose

The general method grows from the earlier paradox audit, the five pillars of Geofinitism, Part I's representational architecture and Part II's finite accounts. The terminology has developed as new distinctions became necessary. The audit is a continuation of that work rather than a fresh conditionless template.

Begin with a construction and a purpose. A word such as number, proof or model is too broad to determine the audit on its own. Identify the particular inscription, report or operation and the action it is being asked to support. This gives the inquiry a local closure question.

The audit may inspect an internal formal operation, an implementation, an FSM translation or a measurement bridge. It can follow several of these together, provided their roles are declared. The report states which one supplies the present question and which supporting paths must be reopened.

16.2 Locate and unfold

Locate the available representation and records. Identify its carrier, the conditions of relevant distinctions, uncertainty, accessible history and licence. A report can point to companion records rather than reproduce them in full. Their accessibility and status belong to the account.

Unfold the relevant FSTs. Follow the registrations, definitions, transformations, exclusions and dependencies that made the present construction available. Distinguish a surviving original record from a reconstruction through the current handler. A reconstructed route can support a new argument while remaining a new event.

At each compression, ask what became portable. A value may replace a ledger, a name may replace a construction, and a theorem may replace a proof. Identify which distinctions were retained, which remain elsewhere, and which are not available. The audit should describe these states before deciding whether any one constitutes a debt for the named task.

16.3 Inspect licences and correspondences

The licence records the commitments permitting the operations. Some are explicit in axioms or protocols; others are supplied through conventions and training. The audit identifies those that matter to the continuation. It need not reproduce every institutional event that helped stabilise them.

A correspondence is then inspected through its source, target, mapping and required preservation. Ask whether the next operation needs a value, membership relation, order, construction history, reversible encoding or another stated feature. The map's advertised strength and the operation's actual need must be matched.

At an empirical crossing, identify the registrations and comparison conditions. At an endogenous bridge, identify the formal domains and relations. The same general procedure follows both while preserving their different licences. A result from one route can then be carried forward with an inspectable account of the transition.

16.4 Witness insufficiency or establish bounded support

Part II's witness remains the central finite instrument. Two exhibited ledgers can have

$$\text{val}_C(L_A) = \text{val}_C(L_B) = v \quad (16.1)$$

while

$$\text{Adm}_C(T_{a,C}, L_A) \neq \text{Adm}_C(T_{a,C}, L_B). \quad (16.2)$$

The report must state a , C , the extracted value rule and the two ledgers. A second form admits both actions but shows that their results fail a named correspondence test. The test and its tolerance are part of that witness.

For bounded sufficiency, identify the examined family and proposed continuations. A finite enumeration can establish support for all its listed cases when those cases exhaust the stipulated family. A different construction can establish support through a formal argument within its declared domain. The audit states which method was used.

An observed agreement in a few cases does not silently become a licence covering unexamined cases. Equally, the absence of complete provenance does not establish that every operation is unsupported. The report follows the support needed for the task and closes at its declared boundary.

16.5 Record debt, cost and continuation

A debt entry names the missing distinction needed by the proposed claim. Record where it was last accessible, its present status and a possible route of recovery. A source file can close one debt; an independent comparison can supply another constraint; a new registration can create a distinction not previously available.

Cost records the work of carrying or extending support. Use the quantities appropriate to the episode. Storage, operations, instrument time and attention can be compared separately. A combined cost requires a declared rule. Debt and cost remain different entries because obtaining a missing distinction and naming its absence perform different jobs.

The audit closes with what has been established, under which conditions, and what could extend the account. Its conclusion becomes another finite representation. The method can then be applied to a paradox, a scientific noun, a statistical claim or an LLM-produced proposal by identifying that construction's own task and relevant trajectory.

The Audit Outcome as a Finite Report

17.1 The findings carried

An audit's outcome is a coherent finite document. It can include a local yes-or-no admissibility judgement, but it also carries the conditions and evidence making that judgement available. A statement that an action is unsupported should identify the particular missing relationship.

The report records function as well as limits. A formal construction can be internally usable, an implementation can support a bounded operation, and a measurement bridge can be recurrent under named conditions. These findings answer different questions. They can be carried together without being reduced to one global verdict.

The conclusion should be readable by a later handler. It states the construction, its task, relevant records, licence, compressions, correspondences, findings and scope. A compact audit card can point to the longer document. The card's own compression should retain the route to any evidence a later continuation will require.

17.2 Different states of missing information

An omitted field in a travelling summary may remain present in a source report. A displaced entry may remain identifiable in another ledger column. A distinction below an earlier registration condition may never have been recorded. A proposed history may be reconstructed from other constraints. These cases need different remedies.

The audit records their status explicitly. A surviving record can be consulted; a reconstruction can be tested through additional relationships; a never-registered distinction requires a new route if one is possible. Confidence in the final numeral does not by itself resolve uncertainty about its history.

This is why representational debt is kept as an ordered record of obligations. Its entries can be closed, altered or left open. A scalar deficit would conceal the difference between retrieving a file and making a new measurement. Their resource demands and evidential functions differ.

17.3 Recompression and the next reader

After unfolding, the report can be compressed again. A scope statement, an audit card and accessible dependencies permit the construction to remain mobile. The audit therefore ends with a usable bead rather than an injunction to carry every page of history at all times.

A later task can reopen that bead. If it needs a distinction not preserved in the present closure, the report can be extended. The new event belongs to the later account and need not be projected backwards into the original registration.

The audit's own report consequently exhibits the theory it uses. It has a finite carrier, a selected history, a purpose and a region of support. Its authority travels through those relationships, and its next continuation remains available for inspection.

Worked Audit I: Infinity

18.1 The claim and the two uses

The specimen is a written inductive-set statement together with a finite successor trace. Its first purpose is to make a classical permission available for a formal argument. Its second proposed purpose is to authorise a particular next step in a finite construction. The audit keeps these purposes separate.

Adopt the formal abbreviation $S(x) = x \cup \{x\}$. A finite displayed trace begins

$$n_0 = \emptyset, \quad n_1 = S(n_0) = \{\emptyset\}, \quad n_2 = S(n_1) = \{\emptyset, \{\emptyset\}\}. \quad (18.1)$$

A further displayed stage can be constructed through the same rule. The trace has finite length and each expression has a finite carrier. The rule and earlier permissions support these formal transformations.

The inductive-set assertion adds its own classical commitment. The audit can follow a formal implication from that assertion without claiming that the trace has produced every member of its witness. The trace is an example of particular stages; the assertion is a permission concerning the formal domain.

18.2 The finite continuation examined

Stipulate a finite episode whose admitted register can store three stage identifiers and their references. The present ledger contains n_0, n_1, n_2 . Its next action is to store a distinct fourth identifier under a rule forbidding overwritten entries and requiring all four to remain accessible in that register.

Under this stipulated rule, the current register does not support the action. The successor expression can still be formed elsewhere; a larger register or another report can extend the account. The unavailable continuation concerns the declared carrier and preservation conditions. It does not follow from a contradiction in the successor rule.

This is a cost and support question with an inspectable boundary. The classical permission can remain available for its formal purpose. The finite episode requires a changed carrier, a changed preservation licence or a new linked record to perform the requested action. The audit records which change is actually made.

18.3 The resulting profile

The construction's function is to supply a formal inductive-set permission and a finite example of successor use. Its representation is the present report and its trace. Its licence includes the chosen set-theoretic abbreviations and the separate finite register rule. The attempted storage continuation is unsupported by the stipulated three-place carrier.

The debt is specific: a fourth accessible stage location is needed under the no-overwrite requirement. A new linked record can discharge it. Naming the location does not create its storage, and enlarging the carrier is an additional event in the ledger.

The case does not identify the Greene–Pascal contribution with the three-place limit. It provides one bounded example of support needed for a continuation. A further Greene–Pascal investigation could compare alternative reports and handlers that hold the same successor relation under stated resource conditions.

Worked Audit II: A Set and Its Construction

19.1 A shared membership presentation

Let a and b be two distinct formal labels. A stipulated admission procedure records their arrival in an ordered ledger and supplies the unordered membership presentation as its portable output. The two accessible ledgers are

$$L_A : \quad a \longrightarrow b, \quad L_B : \quad b \longrightarrow a. \quad (19.1)$$

Under the extraction rule V ,

$$V(L_A) = V(L_B) = \{a, b\}. \quad (19.2)$$

The displayed equality is exact within the chosen formal domain. Both histories are explicitly available in the audit report.

The first task is to decide whether a belongs to the resulting set. The extracted presentation supports the same affirmative result for both ledgers. If this is the only continuation required, the membership compression is sufficient for the stated two-case exercise.

A second task asks the procedure to undo the last admission and report the retained membership. This action is specified by the ordered ledger. It is not specified by an arbitrary removal rule on the unordered set.

19.2 The continuation that separates

Write U for this rollback operation. The two results are

$$V(U(L_A)) = \{a\}, \quad V(U(L_B)) = \{b\}. \quad (19.3)$$

Both operations are admitted. Their resulting membership presentations fail the comparison that would require them to be the same. This is the second form of Part II's insufficiency witness.

A reader supplied only $\{a, b\}$ cannot determine which result the named rollback must produce for these two histories. The missing distinction is the last-admission relationship. A value map that removes that relationship is sufficient for the first task and insufficient for the second.

The witness is entirely finite. It exhibits two ledgers, one extraction, one action and one comparison. It does not require a completed collection of every possible construction history. Nor does it require asserting that sets cannot encode order. An ordered-pair or sequence encoding can carry the needed distinction if it is included in the travelling report.

19.3 The amended report

The report can be extended by attaching the last-admission label or a link to the ordered ledger. Either route supplies support for the rollback in this two-case construction. The audit can then record a new sufficient compression for the named task.

If the labels stand for registered specimens, another bridge is required. The formal rollback identifies a label. Physically retrieving the associated specimen requires its registration, location and identification conditions. The amended symbolic report supplies part of that route, not the whole physical event.

The case therefore closes with two functions and two scopes. Membership is supported by the compact set presentation. Rollback requires an ordering distinction retained in a richer report. Physical retrieval requires a further registered correspondence. Each extension can be followed as another FST.

Worked Audit III: A Polynomial Bridge and Proof

20.1 The two representations

Use formal polynomials over the rational numbers and finite non-empty coefficient lists. Entries in this case are exact formal values under the classical licence; their occurrence in the report is finite. No measured-number object is claimed from the use of rational coefficients.

For $a = (a_0, \dots, a_m)$, define

$$\Phi(a) = \sum_{i=0}^m a_i X^i. \quad (20.1)$$

Here X is a formal indeterminate. Equality of polynomials is coefficient equality under the adopted construction, rather than agreement at only a finite set of evaluation points.

A list is canonical when its last entry is non-zero, except that the zero polynomial uses the single-entry list (0) . Let $\text{can}(a)$ delete trailing zero entries under this convention. Let Ψ read a formal polynomial into its canonical coefficient list. The two return statements are

$$\Phi(\Psi(p)) = p, \quad \Psi(\Phi(a)) = \text{can}(a). \quad (20.2)$$

The second becomes recovery of the original list when that list is canonical. The report thus states its inverse scope at the outset.

20.2 A finite product

Take

$$a = (1, 2), \quad b = (3, -1), \quad (20.3)$$

so that $\Phi(a) = 1 + 2X$ and $\Phi(b) = 3 - X$. The coefficient product is constructed through finite convolution:

$$c_k = \sum_{\substack{0 \leq i \leq m, \ 0 \leq j \leq n \\ i+j=k}} a_i b_j, \quad 0 \leq k \leq m+n. \quad (20.4)$$

All ranges are finite in the present construction. For this specimen, the ledger is

Index	Admitted coefficient operations	Output
0	$1 \cdot 3$	3
1	$1 \cdot (-1) + 2 \cdot 3$	5
2	$2 \cdot (-1)$	-2

Its resulting list is $c = (3, 5, -2)$. The polynomial route gives

$$(1 + 2X)(3 - X) = 3 + 5X - 2X^2. \quad (20.5)$$

The two routes correspond exactly under the formal arithmetic and polynomial conventions.

20.3 The proof unfolded

The operation-preservation statement for the chosen finite lists follows by distributing the two finite sums and gathering terms with the same exponent:

$$\begin{aligned}
 \Phi(a)\Phi(b) &= \left(\sum_{i=0}^m a_i X^i \right) \left(\sum_{j=0}^n b_j X^j \right) \\
 &= \sum_{i=0}^m \sum_{j=0}^n a_i b_j X^{i+j} \\
 &= \sum_{k=0}^{m+n} \left(\sum_{\substack{0 \leq i \leq m, 0 \leq j \leq n \\ i+j=k}} a_i b_j \right) X^k \\
 &= \Phi(c).
 \end{aligned} \tag{20.6}$$

The proof report invokes distributivity, finite reindexing and the conventions for formal powers and polynomial equality. These dependencies are sufficient for the stated preservation claim. Its scope is multiplication in these formal representations.

The audit can unfold the product one coefficient operation at a time, as the table does for the example. It can then recompress the path into the preservation statement. A later formal calculation can use that statement without repeating every displayed multiplication.

If the coefficient arithmetic is implemented with rounding, the implementation becomes another trajectory. Its relation to the formal proof requires the declared arithmetic regime. The exact preservation statement is not silently reassigned to an arbitrary device procedure.

20.4 The distinction left outside the polynomial

Now consider two raw list records,

$$r_A = (1, 2), \quad r_B = (1, 2, 0). \tag{20.7}$$

They are admissible raw records before canonicalisation. Their polynomial outputs coincide:

$$\Phi(r_A) = \Phi(r_B) = 1 + 2X. \tag{20.8}$$

The third zero in the second record supplies no additional coefficient content to that polynomial. It does, however, occupy a slot in a stipulated storage account.

Define the next action as writing a specified value into an already allocated third slot, under a rule forbidding extension of the record. The action is not admitted for the two-slot record and is admitted for the three-slot record. Their shared polynomial is insufficient to decide this storage continuation.

The witness concerns allocated record length. The polynomial bridge preserves the formal relationship it advertised, including the product just proved. The omitted length becomes a debt only when the storage action is proposed. A report carrying the raw length beside the polynomial can support the action. The bridge's algebraic success and the later storage insufficiency are both retained in the profile.

20.5 The bridge report

The audited construction names source and target representations, the forward map, canonical return map, multiplication correspondence and raw-record convention. Its function is exact formal mobility between coefficient work and polynomial work. Its return operation recovers canonical coefficient content.

The original raw carrier and its full history are not recovered by the canonical return statement. A continuation needing allocated length or provenance requires another entry. This is an explicit instance of the general bridge inquiry: what has been preserved, and is that preservation enough for the next task?

The report can now travel as a finite bead. It carries a useful theorem and a named boundary of its use. Recompression has retained the function while making the particular insufficiency inspectable.

Worked Audit IV: A Registered Partition

21.1 The proposed physical claim

The specimen is a proposed report of partitioning a registered whole and retaining half. It develops Part II's arithmetic story into an empirical bridge scenario. The numerical examples below are stipulated descriptors; they are not presented as a newly performed measurement.

The whole requires a reference, carrier and registration procedure. The partition requires an action, an account of retained and set-aside portions, and any relevant residue or untracked disposition. The value report can then be extracted under a declared rule. The classical expression $1 - \frac{1}{2} = \frac{1}{2}$ supplies a value relation; the empirical account needs the additional trajectories.

For a particular partition, write schematically

$$L : W \longrightarrow (R, X; h), \quad (21.1)$$

where W is the registered whole, R the retained portion, X the set-aside portion and h the accessible action record. A residue or an unresolved destination becomes a further entry when the episode requires it. These symbols name positions in a finite report.

21.2 The bridge to a ratio

A numerical ratio needs compatible reference and unit conventions. If a measured retained extent is compared with the whole, the extraction

$$v = \frac{\ell_R}{\ell_W} \quad (21.2)$$

is a formal operation on the supplied values. Its empirical use depends on their registration conditions and on the comparison that admits the result as half within a stated regime.

The claim of an exact arithmetic half and a measured half have different licences. The latter must state its comparison criterion, including relevant uncertainty and correlations in the measurements. A ratio alone does not decide whether the two lengths were measured independently, share a calibration, or were inferred through another construction.

An audit can therefore inspect the reference trajectory, the partition trajectory, the retained and set-aside registrations, the ratio extraction and the later claim. A compact remainder can be sufficient for one decision while being insufficient for custody, identification or a further physical operation.

21.3 A finite continuation witness

Retain Part II's stipulated eight-unit whole with four unit beads and two double-unit beads. Under a rule allowing selection of whole admitted beads and forbidding their division within

the exercise, the two ledger paths are

$$\begin{aligned} L_A : (1, 1, 1, 1, 2, 2) &\longrightarrow (1, 1, 1, 1) \longrightarrow (1, 1), \\ L_B : (1, 1, 1, 1, 2, 2) &\longrightarrow (2, 2) \longrightarrow (2). \end{aligned} \tag{21.3}$$

The notation records extent descriptors; the fuller ledger gives bead identities and set-aside entries. Both retained states report two of the original eight units, or one quarter.

The proposed next action is another halving under the same whole-bead rule. It is admitted on L_A and unavailable on L_B . A refined operation could create new distinctions within the double-unit bead, but that refinement would change the construction and require an additional ledger event.

The witness remains within the stipulated symbolic exercise. Applying it to a physical partition requires evidence that the stated registration and operation rules describe the particular apparatus and material episode. The audit records that bridge requirement rather than treating a bead drawing as a measurement.

21.4 The spectral continuation

Part II's spectral comparison provides a longer scientific route to which the same method can travel. A reference registration and an observing registration are transformed into comparable reported feature values. Their ratio or difference becomes a portable parameter, while calibration, identification, exclusions and uncertainty remain in supporting records.

A later interpretation needs the correspondences relevant to that claim. The absence of the full account from the parameter does not alone determine the physical mechanism of the recorded difference. The audit can identify which record or new registration would support a proposed continuation.

The partition and spectral cases therefore share an audit architecture, while their physical trajectories remain distinct. Each requires its own registrations and comparison rules. The present volume supplies the questions and finite report structure with which those routes can be followed.

The Audit Audits Itself

22.1 Terms as compressed trajectories

The language of this volume performs the same operations it inspects. FST, bead, licence, bridge, Alphonic and finite report gather earlier trajectories into portable expressions. Their definitions and examples make some of that support accessible. Their use should be examined when a new sentence asks them to carry another relationship.

A terminology ledger therefore records the established use, source trajectory and present extension of a term. Functional remains the word represented by the F in FST. A report remains a coherent document; a value remains an extraction; a ledger remains an accessible record of actions. Their short names must not remove these distinctions.

Greene–Pascal is explicitly provisional. Its use here points to representational support for holding an Alphonic distinction. A later minimum-chain theorem, cost rule or empirical relation would require additional construction. Those possibilities are retained as continuation questions, not supplied retroactively by the name.

22.2 The supporting sources

The source record has several roles. Parts I and II supply the commitments and local witnesses carried forward. The planning conversation supplies the provenance of the volume’s organisation. The historical essay supplies an interpretative trajectory concerning licensing. Teaching texts supply checks on the conventional formal presentation.

These roles should not be merged. A source for a ZFC formula does not supply the FSM audit of it. A historical interpretation does not prove the quarter-bead witness. The witness does not establish a general law of spectral change. Each dependency is used for the relationship its report makes available.

The bibliography is consequently a finite source direction, not a claim that the wider corpus has been exhaustively audited. A later scholarly edition can extend the historical record and compare other formal presentations. The present draft identifies the construction it actually follows.

22.3 Working with a language model

An LLM can help unfold a proposal, reconstruct a dependency, suggest a term or compose a report. Its output is another finite trajectory produced under a prompt, context and system. The audit can identify which parts were inherited, reconstructed or newly proposed.

A fluent continuation can introduce a changed meaning while retaining familiar words. In the present project, shifting Greene–Pascal from representational support to a resource

threshold would be such a change. Recording the distinction reduces the risk that a convenient restatement quietly redirects the programme.

The practical method is to retain the user's committed terms, the sources consulted, the proposed extension and the status of each new relationship. A novel claim can remain open for exploration while its supporting trajectory is made inspectable. This extends the mathematical audit into the wider work of developing ideas.

22.4 The stopping condition

An audit can itself become too elaborate for its purpose. Each reopened dependency creates further possible paths. The stopping condition therefore belongs to the licence of the present task. It states which relationships have been examined and which remain referenced, reconstructed or unresolved.

The report can still be useful at that point. A finite witness, a clarified permission or a recoverable source link can support the intended continuation. A later use may reopen another dependency. That event creates a new account.

Self-audit therefore preserves the dynamic character of FSM. Its terminology and methods remain situated in a continuing symbolic lattice. The report has an author, carrier, history and function, and its own next trajectory remains available.

Conclusion: Unfold, Audit and Carry Forward

Part I followed how a representation becomes available. Part II followed how a finite report carries an account and how its extracted values can become insufficient for a named continuation. Part III has followed inherited symbolic constructions through those relationships. The formal beginning has been reopened towards the trajectory that holds it.

The ZFC specimens show a differentiated architecture of permissions. Identity, collection, mapping, selection and closure support different mathematical functions. Their finite uses can be inspected, their supporting statements made accessible, and the requirements of a proposed FSM translation identified. The audit preserves the distinction between a finite inscription and the commitments expressed through it.

The worked cases make the method operational. A successor trace and a storage rule identify a particular unsupported continuation. A set presentation supports membership while an ordered rollback needs another account. A polynomial bridge preserves exact formal multiplication and canonical return while a storage operation needs raw-record length. A partition report connects arithmetic values to the further conditions of registered action.

These findings are finite and local. Their force comes from the exhibited trajectories, declared operations and stated comparisons. The audit can also establish bounded support. It does not require every compact result to carry the same debt, nor every longer report to be functionally preferable.

The Greene–Pascal framing connects distinction to the trajectories needed to hold and use it. Its development can proceed through questions of minimum sufficient support, handler dependence, compression and cost. This volume has located those questions without turning their provisional name into a completed calculus.

The mountain remains a useful image. A refined endpoint can organise a further explanation, while its producing trajectory remains available to inspection. The audit follows both directions. It asks what the endpoint enables and what its new use requires from the path that made it present.

The separate Mathematical Objects programme can now receive a constructive handover. A proposed FSM object should disclose its carrier and construction, operative distinctions, uncertainty, accessible provenance, licence, transformations and permitted continuations. Its relationships with classical objects should be specified through declared correspondences. A return operation should state what it reconstructs and under which conditions.

The audit is itself one such finite construction. It can travel, be shortened, reopened or extended. Its function is to make another trajectory inspectable enough for the work being proposed. From that finite place, another construction can begin.

APPENDIX A

A Reusable Audit Card

A card is a compact entry point into a longer finite report. Its fields are selected for the task. The following form gathers the present method without requiring every case to possess identical records.

Field	Entry to be carried
Construction	The particular claim, object, inscription or operation being inspected.
Purpose	The action, decision or correspondence it is being asked to support.
Representation	Carrier, version or occurrence, accessible companion records and relevant distinction conditions.
Licence	Formal rules, operational protocol, admission conventions and declared scope.
Trajectory	Registrations, transformations, dependencies and selected unfolding depth.
Compression	What became portable; what was retained, moved elsewhere, reconstructed or unavailable.
Correspondence	Source, target, mapping, preserved relationships, direction and return conditions.
Continuation	The named next operation and the conditions of its admissibility.
Witness or support	Exhibited cases, a bounded check or a formal argument under a declared domain.
Debt	The particular required distinction, its last accessible stage, present status and possible remedy.
Cost	Declared quantities for carrying, checking, recovering or refining support.
Scope	What the present audit establishes and where its examined support closes.
Next event	A record to retrieve, dependency to reopen, reconstruction to test or registration to make.

For the polynomial case, the construction is the map from raw coefficient lists to formal polynomials. The formal multiplication purpose is supported by finite distributive reindexing. Canonical coefficient recovery is supported by the stated convention. Writing into an already allocated third slot requires raw length, which the polynomial alone does not carry. The debt entry therefore names that length field rather than an unspecified loss of information.

A card can record an unresolved result. It should distinguish absence of a checked finding from a checked finding of unsupported continuation. Those entries can require different next events. The compact card remains functional when its scope sentence makes this status available to another reader.

APPENDIX B

Terminology and Notation Ledger

Functional Symbolic Trajectory

A finite pathway of symbolic states and transformations through which constraints, uncertainty, meaning and admissibility can be carried. Functional identifies its capacity to alter available continuations. Its written linear sequence is one representation of the process.

Finite report

A bounded, instantiated coherent document gathering one or more FSTs and their admitted connections for a declared purpose. It can include tables, proofs, images, references or spoken accounts. Its coherence and scope are part of its construction.

Ledger

An accessible finite record of relevant actions and distinctions. It can retain ordering, disposition and conditions omitted from a compact output. It is selective and does not claim a completed history.

Reported value

A particular extraction from a ledger or report under a rule. Its compactness can support mobility. Whether it is sufficient depends on the continuation it is asked to support.

Bead

An admitted finite distinction that can occupy a position in a symbolic chain and condition an operation. Inscriptional marks, counted members and registered extent descriptors require separately declared roles.

Alphonic distinction

The operative condition at which a relevant difference becomes non-single in a registration or representation. A report may contain several such conditions. A proposed universal physical limit is a further inquiry.

Greene–Pascal

In this draft, a provisional framing of the FST or supporting region of FSTs needed to hold and functionally use an Alphonic distinction. Minimum sufficient support, handler dependence and quantitative cost remain development questions.

Grand Corpus

A pointer to the wider accumulated and continuing field of symbolic trajectories through which local expressions are sustained. A finite audit accesses a bounded region; no completed inventory of every possible trajectory is claimed.

Handler

The current endogenous lattice through which a representation is reconstructed, compared and incorporated into a continuation. Different handlers can reconstruct corresponding relationships through different histories.

Licence

The commitments and permissions under which a construction or transition is admitted. A formal axiom, an encoding convention and an operational protocol can each contribute to a licence, with their roles stated.

Bridge

A declared correspondence between constructions or trajectories. An endogenous mathematical bridge and an empirical basin crossing have different conditions. Direction, preservation and return operations belong to its account.

Conditional insufficiency

A witnessed failure of a compact representation to support a named continuation under declared conditions. The witness can concern different admissibility judgements or results failing a specified comparison.

Representational debt

The particular distinction or record owed by a claim when the presented support does not carry it. Debt entries distinguish recoverable records, reconstructions and never-registered distinctions.

Representation cost

Declared resources used to produce, carry, inspect, recover or refine support. Quantities with different units remain separate unless a conversion rule is supplied.

Symbolic drag

The residual effect of an earlier trajectory and its commitments upon a compressed expression in a later use. Unfolding can make some of that dependence accessible.

Conditioned correspondence

A comparison under stated licence, distinction, uncertainty and history conditions, written with the interactional bar. It need not identify carriers or complete trajectories.

ZFC Presentation and Dependencies

C.1 Grammar and closure

The presentation in Chapter 6 uses classical first-order logic with equality. Its sole non-logical relation is membership. Variables range over sets. The symbols for empty set, subset, union, intersection, ordered pair, function and domain are abbreviations developed through the stated permissions.

All displayed axiom instances are universally closed over their unbound parameters. For Separation, $\varphi(x, \vec{p})$ has no free occurrence of the target variable B . For Replacement, $\theta(x, y, \vec{p})$ likewise excludes B , and its output uniqueness is required for each input in the source set. Additional bound variables internal to these formulas are allowed.

A schema is described in the metatheory. The report's pattern for formulas is finite; the classical schema supplies an instance for each formula meeting its conditions. A particular finite derivation records the instances it invokes. This distinction is part of the provenance of the chosen formal use.

C.2 A deliberately redundant list

The chapter displays Extensionality, Empty Set, Pairing, Union, Power Set, Infinity, Separation, Replacement, Foundation and Choice. The separate empty-set assertion and retention of both schemas are intended to expose their different functions. The list is not a claim of independence of every displayed entry.

Empty-set existence can be obtained by a Separation instance from an available set. Infinity supplies such a source within the selected presentation. Extensionality establishes uniqueness. Further redundancies can be examined in other presentations, but their proofs are not required to use the present audit specimen.

The abbreviations in Infinity depend on the relevant empty-set, pairing and union constructions. The least inductive set requires a further construction beyond selecting an arbitrary inductive witness. The audit therefore retains the difference between a witness named by an existence permission and an object selected by another definition.

C.3 Containment and scope

The set-centred comparison does not posit a universal set inside ZFC. To see the obstruction in the present permissions, suppose a set U contains every set. Separation gives

$$R = \{x \in U : x \notin x\}. \quad (\text{C.1})$$

Because R would itself belong to U , the defining relationship would yield $R \in R$ exactly when $R \notin R$. The classical logic therefore excludes the supposed universal set under these

assumptions. This finite derivation records why a broad organising language of sets should not be confused with a literal master set.

A class expression can instead serve as a syntactic abbreviation for a property in an appropriate metatheoretic discussion. Its use does not automatically assert a further set. FSM's Grand Corpus likewise has its own declared role as a pointer to continuing symbolic support; it is not introduced as such a universal set.

The conventional presentation has been checked against the teaching sources listed in the bibliography. Their role is to ground the mathematical specimen. The Geofinite reading of licences, compression and continuation is developed in the present report.

APPENDIX D

Prior Explorations and Provenance

Part I established the generonic boundary, the three basins, conditioned representation and the compression–decompression account. Part II developed the bead, ledger and classical paths, the quarter-bead witness, debt entries and a finite audit card. Those reports are the principal dependencies of the present construction.

The earlier paradox audit, described in the planning discussion, supplies a programme lineage. Its questions concerning finite distinction, uncertainty, provenance and admissibility are extended here through the more developed language of reports and continuations. This appendix records that relationship without claiming to reproduce an unavailable complete earlier audit.

The Part III planning conversation added the comparison between sets and FSTs as wrappers, the trajectory supporting an axiom, bridges between mathematical fields, the mountain image, functionality and the need to audit new terminology. The revised outline of 5 October 2026 arranged those contributions into the present sequence.

The subsequent alignment clarified two choices. Greene–Pascal begins with representational support for an Alphonic distinction and remains a framing open to development. Part III performs the audit, while fuller object reconstruction continues in the separate Mathematical Objects programme. The polynomial and partition cases were then selected as concrete specimens for the method.

The chapter prose, exact examples and finite proof traces in this working draft are constructions developed from that outline. They remain available for revision. The source trajectory, agreed scope and proposed developments have been kept distinct so that the next reading can inspect where each entered.

Greene–Pascal Development Questions

The present framing asks what supporting trajectory holds a distinction sufficiently for its named use. A first development route would choose a bounded operation and compare reports that make it reconstructable. The comparison needs a specified handler or an explicit family of handlers, a correspondence test and a stopping condition.

A proposed minimum sufficient account should state what counts as part of the support. If a short label relies on a trained reader's extensive history, counting only the visible characters omits a relevant part of the construction. If a self-contained report carries more of that history explicitly, its greater length can support another reader. The minimum is therefore conditioned by the handler and task.

A second route would vary the operative distinction condition. It would ask which new support becomes necessary when a finer difference is admitted, and which earlier compressions remain sufficient. An instrument change can alter the relationship, so the comparison should record which apparatus, procedure and purpose are held fixed.

A third route would compare compression schemes. Two reports could support the same continuation while using different storage, reading or checking resources. A finite witness could identify a further compression at which support fails. The resulting account would connect cost to function without introducing a universal cost unit.

A fourth route concerns persistence across handlers and contexts. A distinction may be held locally but difficult to communicate. A richer supporting region could make it available across several readers or systems. The comparison would follow their reconstructed trajectories and the relationships required by the task, rather than assuming identical internal states.

These routes are compatible with the wider questions of continuance raised in the conversation. Their present value is to name constructive investigations. They do not yet supply an absolute minimum chain, universal physical boundary or scalar law. A future result would enter as another finite report with its own registrations, rules and source record.

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End of Part III